

ARTIFICIAL INTELLIGENCE & NATURAL RESOURCE CONSUMPTION

A structured report on electricity, water, materials, infrastructure and waste

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1. Introduction

Artificial Intelligence (AI) is increasingly used for search, recommendation, language processing, image generation, scientific research, automation and decision support. Although AI is experienced mainly as software, it depends on physical infrastructure such as servers, accelerators, storage, networks, cooling systems and manufacturing supply chains.

The resource footprint of AI therefore extends beyond electricity used by a model. It can include water used directly or indirectly, minerals and industrial materials required for electronic equipment, land and construction materials for data centers, and waste generated when hardware is replaced. The scale varies with model size, workload, hardware efficiency, data-center design, location and electricity source.

2. Natural Resources Involved

Resource	AI Use	Why Needed	Main Concern
Electricity	Training, inference, storage, networking, cooling	Powers computation and infrastructure	Energy demand and emissions
Water	Cooling and upstream electricity generation	Supports thermal management and power systems	Local water stress
Minerals & materials	Chips, servers, cables, batteries and equipment	Required by electronic hardware	Mining and supply-chain impacts
Land & construction	Data centers and supporting infrastructure	Physical facilities are necessary	Land and construction footprint
Hardware	GPUs/accelerators, CPUs, memory, storage	Performs AI computation	Manufacturing and replacement
E-waste	Retired servers and components	Hardware has an end-of-life stage	Recovery and disposal

3. AI Resource-Consumption Chain

RAW MATERIALS → HARDWARE → DATA CENTER → TRAINING / INFERENCE → POWER & COOLING → RETIREMENT

4. Why It Matters

- AI services can operate at very large scale, so resource requirements per task can become significant when multiplied across many users and workloads.
- Impacts differ by region because electricity mix, cooling technology, climate, water availability and supply chains are different.
- Efficiency improvements can reduce resource use per computation, but rapid growth in AI demand can offset part of those gains.
- Responsible AI development should consider efficiency and infrastructure impacts in addition to accuracy, usefulness and safety.

AI RESOURCE CONSUMPTION — KEY AREAS

Electricity • Water • Materials • Data Centers • E-Waste

5. Electricity Consumption

Electricity is a central resource for AI because computational workloads run on CPUs, GPUs and other accelerators. Power is used for training, fine-tuning, inference, data movement, storage and cooling. There is no single universal energy value for an AI task because requirements depend on hardware, model architecture, workload size, utilization and facility efficiency.

- Training can require substantial computation because models process large datasets repeatedly.
- Inference occurs whenever a model generates a response or prediction; at high usage volumes, inference can become a major operational workload.
- Cooling and supporting infrastructure add to the electricity requirement of computing equipment.
- The environmental effect of electricity depends partly on how it is generated; lower-carbon electricity can reduce associated greenhouse-gas emissions.

6. Water Consumption

Water can be associated with AI infrastructure through direct data-center cooling and indirect water use connected with electricity generation. The amount varies with cooling design, climate, facility operation and the local electricity system. Water consumption should therefore be evaluated at the facility and regional level rather than treated as one fixed value for every AI query.

7. Materials, Minerals and Hardware

AI computing depends on sophisticated electronic hardware. Semiconductor manufacturing, servers, memory, networking equipment, power systems and cooling equipment require raw materials and industrial processes. The supply chain may involve mining, refining, manufacturing, transportation and assembly.

- Specialized accelerators increase computational performance but require complex semiconductor manufacturing.
- Servers contain processors, memory, storage, circuit boards, power supplies and networking equipment.
- Rapid growth can increase demand for materials and place pressure on complex global supply chains.
- Longer hardware life, repair, reuse and recycling can reduce material demand.

8. Data Centers and Physical Infrastructure

AI requires physical data centers with reliable power, networking, security, cooling and environmental controls. Constructing or expanding facilities requires land, concrete, steel, electrical equipment, cables and other materials. Large new computing facilities can also increase demand for grid connections and supporting energy infrastructure.

9. E-Waste and Hardware Lifecycle

AI hardware follows a lifecycle: extraction → manufacturing → deployment → operation → upgrade or reuse → retirement. Rapid technology changes can shorten the practical replacement cycle of some equipment. Responsible management should therefore consider refurbishment, component recovery, recycling and safe disposal.

REDUCING AI'S RESOURCE FOOTPRINT

Efficiency, responsible infrastructure and lifecycle thinking

10. Strategies for More Sustainable AI

Area	Possible Approach	Potential Benefit
Model efficiency	Use smaller capable models; optimize architectures; avoid unnecessary computation	Lower compute and energy requirements
Hardware efficiency	Use efficient accelerators and improve utilization	More computation per unit of energy
Data-center design	Improve cooling, power management and facility efficiency	Reduced operational resource use
Electricity	Increase low-carbon electricity where feasible	Lower associated carbon emissions
Water management	Use cooling suited to local climate and water availability	Reduce local water pressure
Hardware lifecycle	Repair, reuse, refurbish and recycle equipment	Reduce material demand and e-waste
Workload management	Match model and computing resources to the actual task	Avoid unnecessary computation
Measurement	Track energy, water and lifecycle indicators	Enable evidence-based improvement

11. Role of Developers and Organizations

- Choose the smallest capable model for the task instead of assuming the largest model is always necessary.
- Optimize inference, caching, batching and data pipelines to avoid redundant computation.
- Measure resource use where practical and include environmental indicators in infrastructure decisions.
- Consider location carefully where water availability or grid capacity is constrained.
- Plan hardware procurement around utilization, useful life, reuse and end-of-life management.
- Distinguish measured environmental data from estimates when communicating sustainability claims.

12. Key Challenges

- There is no single universal resource-consumption value for one AI query; workloads and infrastructure differ.
- Resource impacts can shift between categories, creating energy-water or other trade-offs.
- Rapid AI growth makes long-term infrastructure and electricity planning increasingly important.
- Supply-chain impacts are difficult to measure completely because hardware production spans many companies and countries.
- Environmental performance requires lifecycle thinking rather than measuring only model-training electricity.

13. Conclusion

AI is a digital technology built on a physical resource base. Its natural-resource footprint includes electricity, water, materials, land, infrastructure and hardware. A useful approach is to measure where resources are consumed and improve efficiency across the full lifecycle. Smaller and better-targeted models, efficient hardware, responsible data-center design, appropriate cooling, cleaner electricity and hardware reuse can all contribute to reducing resource intensity.

14. Selected References / Further Reading

- International Energy Agency (IEA), Energy and AI — analysis of AI-related electricity demand and data-center energy use.
- U.S. Department of Energy — data-center electricity use and energy-efficiency resources.
- Masanet et al., Science (2020), Recalibrating global data center energy-use estimates.
- Li et al., Making AI Less 'Thirsty' — discussion of the water footprint of AI models.
- International Telecommunication Union (ITU) — e-waste and sustainable digital infrastructure resources.